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E-waste management awareness and intentions among youth consumers: the mediating role of motivation

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Abstract: The paper empirically explores the role of general awareness, motivation and intentions in e-waste. It proposes an original framework to explore the influence of e-waste awareness on e-waste management intentions of consumers through the mediating role of e-waste management motivation. The data collected from young users were analysed using mediation model proposed by Preacher and Hayes (2008). The questionnaire was administered to study the identified pertinent issues in e-waste management and the study found that e-waste general awareness of users significantly affects the e-waste management intentions through motivation of the users for their better health and environment.

Keywords: e-waste awareness; e-waste motivation; e-waste management intentions; EWI; mediation model.

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1 Introduction

The spectacular development in information and communication has undoubtedly brought enormous changes in our economies, industries and concomitantly enhanced our quality of lives. Additionally, the unprecedented growth of ICT industry coupled with high product obsolescence and short product life span (Kang and Schoenung, 2004), posing a great threat to the human health and environment in the form of electronic waste (e-waste). Researchers have been studying the manifold issues in e-waste, particularly in the context of impact of wastage severity on our environment and health (Ni et al., 2010; Murugesan, 2008), the disposal and recycling activities of e-waste (Kahhat et al., 2008; Hirschier et al., 2005) and responsibilities of e-waste management (Khetriwal et al., 2009), however these issues were not straightforwardly focused through the consumers' intentions to manage the problems of e-waste. It is asserted that boarder dimension of e-waste management intentions (EWI) among youth consumers remain under-researched. Despite of scholars' attempt to empirically explore the extent to which, or how, the outgrowing of older ICT leading to e-waste which in turn causing major environmental problems and endangering human health (Robinson, 2009), very few attempts have been done in examining the impact of e-waste awareness on EWI among youth consumers in India. Nnorom and Osibanjo (2008) emphasised on producers' responsibility towards material waste management in the product life cycle, i.e., who will produce the goods should be responsible for the extraction to recycling, reuse and disposal of the goods. Although Bhat and Patil (2013) attempted to study and comprehend the e-waste awareness and highlighted the importance of consumer's awareness and responsibility in handling of e-waste, the management of e-waste through consumer's intentions has been neglected. As such, in line with consumer's awareness in e-waste management, this paper aims to examine its impact on EWI among youth consumers.

Conceptually, management of e-waste includes handling, disposing and recycling pattern of e-waste caused by loosely discarded, surplus, obsolete, broken, electrical or electronic devices. Barr et al. (2001) discussed the three attributes associated with recycling or disposing behaviours patterns in an individual namely environmental values, situational factors and psychological factors and linked the altruistic motives, intrinsic and extrinsic motivations to recycling behaviour. Moreover, it has been found that consumers with environmental values and motivation likely to behave environmentally or sustainably (Sharma and Kesharwani, 2015; Sharma and Sharma, 2014), and their concern for environment intend them to perform activities which includes recycling or sustainable disposing of waste (Schultz et al., 1995 and Simmons and Widmar, 1990). Indeed, Goldenhar and Connell (1993) and Hopper and Nielsen (1991) have studied that conceptual model of recycling behaviour where they explained the role of individual's motive to recycle the goods. Indeed, the degree of these variables has been under-researched with the perspective of e-waste in ICT among youth consumers in

India. Hence the objective of this paper is to explore the impact of e-waste management awareness (EWA) on the EWI among the youth in India towards the disposal and recycle of unused ICT products.

This paper prepared as follows: a literature review presents the association among e-waste awareness, e-waste management's motivation and intentions to manage the e-waste products. Following an overview of research methodology and analysing section using mediation model proposed by Preacher and Hayes (2008). The next section summarises the finding of the study. Conclusions and future scope are drawn in the final section.

2 Literature review and framework

Owing to growing concern at large, Government of India is identifying e-waste as a priority and initiating certain programs to create awareness among the stakeholders (Rajya Sabha Secretariat Secretariat, Research Unit (Laridis), Govt. of India, 2011). Indeed, Hicks et al. (2005) claim that awareness regarding recycling, dumping of old or broken e-products certainly change the attitude of the users. Further, Sharma and Kesharwani (2015) stated that increasing the environmental awareness influences the sustainable behaviour. It is quite possible to state that awareness towards manifold problems generated from e-products intend the users to coordinate their efforts in achieving sustainable development. Hence, in a straightforward and linear fashion, this study claims that e-waste management knowledge/awareness leads to intentions towards management of e-waste among the users.

Hypothesis 1 Users' awareness towards e-waste is positively affects EWI among them.

Motivation embraces the external or internal forces that guide the desire, enthusiasm and determines certain course of action. Conceptually, altruistic, intrinsic and extrinsic motives increase recycling (Hopper and Nielsen, 1991) and sustainable behaviour (Sharma and Sharma, 2014). In the context of sustainable studies, it has been reported that motivational factors positively correlate with environmental behaviour (Sharma and Sharma, 2014a; Bui, 2005). It has been found that consumers are concerned about environmental problems because of its consequences on plants, their lifestyle, their health, children, their children, people of their country, birds and animals, themselves and their future (Sharma and Sharma, 2014b). As such a subjective view of e-waste impact on environment, the present study postulated that there exists an association between user motivation to organise e-waste and EWI.

Hypothesis 2 Users' motivation positively affects EWI.

Terazono et al. (2011) proposed that improved and standardised awareness programs should be initiated by developing countries to motivate the citizens to reduce e-waste. In this context, Wang et al. (2011) suggested four determinants of Beijing residents' willingness and behaviour in e-waste recycling is notable. These are convenience of recycling facilities and service, residential conditions, recycling habit and economic benefits which explained the residents' characteristics of recycling and preference of e-waste recycling pattern. Hence, it is quite possible that e-waste awareness among users motivates them to administer e-waste.

Hypothesis 3 Awareness towards e-waste positively affects users' motives to manage e-wastes.

As we conceptualise awareness effecting motivation and motivation effecting e-waste intentions, it throws up another question; does user's motivation play a mediating or moderating role, if at all, between e-waste awareness and EWI. In case of mediating variables there is a casual relationship between antecedent and consequent variables. Whereas in case of moderating variables, a variable has a contingent effect on antecedent and consequent variables. Motivations anticipate prospective actions, determine of goal and accordingly plan course of action (Bandura, 1989). Therefore, rather than being present or absent (i.e., moderating variable) motivation is tacitly to be present in overt form is forming e-waste intentions. If e-waste awareness presumably affects motivation towards management of e-waste, then motivation is more likely to assume role of a mediator between users' awareness and EWI. Thus,

Hypothesis 4 Motivation has a mediating effect on users' awareness and EWI.

3 Method

3.1 Sample and procedure

The study is based on analysis of data collected through questionnaire/instruments, using Mediation Model proposed by Preacher and Hayes (2008).

3.2 Sample

The unit of analysis in this study is consumer level. The research object of this study focuses on the young ICT consumers in Delhi and NCR. In National Youth Policy Report (2014), 'Youth' was defined a person of age between 15–29 years and they comprise 27.5% of the population. The questionnaires were randomly mailed to one eighty UG and PG students, where 157 participated in the survey. Though the sample was not completely random, yet it was not devoid of it either. A reasonable degree of randomness could be assumed because of absence of systematic effort in the selection the respondents. The convenience bias however cannot be fully ruled out. It might adversely affect the generalisability of the results. However, the limitation is acceptable given the exploratory nature of the study. Prior mailing the questionnaires to the students five area experts and scholars were requested to amend the questionnaire to perform the content validity. Gender wise distribution of the sample showed that 43% were females and 57% were males. 30% respondents were graduates and rest were PG or above.

3.3 Instruments

To study the impact of spirituality on green purchasing intentions data were collected in respect of three variables: e-waste awareness, e-waste management motivation (EWM) and EWI. A questionnaire was administered (Appendix) for the purpose. Questionnaire included ten items measuring the three variables and each item was measured on a five-point Likert scale ranging from 'Strongly agree' to 'Strongly disagree'.

Descriptive statistics comprising means and standard deviations are shown in Table 1 where Cronbach's alpha of each Instrument which is employed to measure their internal

reliability. In the early stage of research Cronbach's alpha 0.6 is acceptable criterion for determining usable reliability of instruments. Cronbach's coefficient estimated for e-waste awareness scale is 0.698. In respect of EWM it is 0.732 and EWI 0.842. Thus, all variables in the present study have a Cronbach's value more than 0.6; hence all the instruments may be deemed to have fairly acceptable/adequate internal consistency.

Table 1 Descriptive statistics

	<i>Descriptive statistics and correlation matrix</i>					
	<i>Mean</i>	<i>Std. deviation</i>	<i>EWA</i>	<i>EWM</i>	<i>EWI</i>	<i>α</i>
EWA	3.671	.714	-			0.698
EWM	3.620	.868	.643**	-		0.732
EWI	4.009	.885	.635**	.644**	-	0.842

Note: $P < 0.01$

4 Data analysis

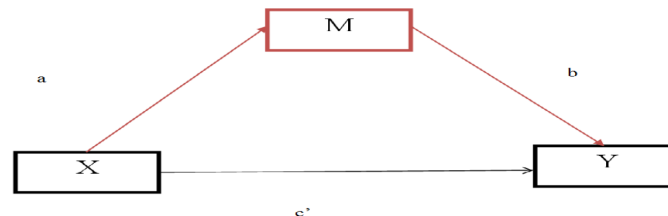
In this study data is analysed by using mediation model which focuses on the estimation of the indirect effect of X on Y through an intermediary mediator variable M causally located between X and Y (i.e., a model of the form $X \rightarrow M \rightarrow Y$) (Preacher and Hayes, 2014), where X is the input variable, Y is output and M is the mediating variable. As per Figure 1, we can observe that X variable exert its effects on Y variable which is commonly intervened by one or two variables denoted by M and this variable has a causal relationship between X and Y. In this casual system at least one casual antecedent X variable is projected as influencing an outcome Y through a single intervening variable M (Preacher and Hayes, 2008). Such model traces two pathways which influences Y by direct effect and indirect effect. In direct effect, pathways lead from X to Y without passing M. In indirect effects, a pathway of X to Y is lead through M. There are two consequent variables forming two equations and these equations can be estimated by conducting OLS regression analyses using SPSS or by using PROCESS.sps in SPSS by Andrew F. Hayes.

$$M = i_1 + aX + e_M \quad (1)$$

$$Y = i_2 + c'X + bM + e_Y \quad (2)$$

Diagrammatically, simple mediation model may be represented as below:

Figure 1 A simple mediation model (see online version for colours)



Source: Preacher and Hayes (2008)

In the present study, we report the confidence interval of these coefficients as well as p-values based on such sampling distribution using PROCESS.SPS software developed by Hayes.

5 Results

Table 2 presents, the results of three OLS regressions estimating the impact of e-waste awareness on EWI (Total Effect), motivation and influence of motivation on management intentions (Indirect Effect), while Figure 2 shows diagrammatic presentation of the model.

Figure 2 Statistical diagram of mediation model of GPI (see online version for colours)

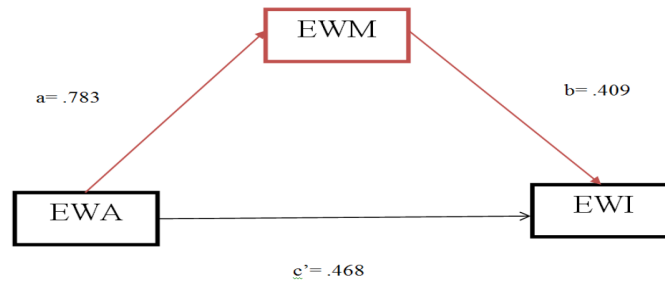


Table 2 Direct, mediation and total effect of EWA and EWI

<i>Consequent: EWM</i>						
<i>Model 1</i>						
<i>R</i>	<i>R-sq.</i>	<i>MSE</i>	<i>F</i>	<i>df1</i>	<i>df2</i>	<i>p</i>
.6432	.4137	.4469	70.5669	1.0000	100.0000	.0000
<i>Variables</i>						
	<i>Coeff.</i>	<i>SE</i>	<i>t</i>	<i>p</i>	<i>LLCI</i>	<i>ULCI</i>
Constant	.7476	.3484	2.1460	.0343	.0565	1.4388
EWA	.7826	.0932	8.4004	.0000	.5977	.9674
<i>Consequent: EWI</i>						
<i>Model 2</i>						
<i>R</i>	<i>R-sq.</i>	<i>MSE</i>	<i>F</i>	<i>df1</i>	<i>df2</i>	<i>p</i>
.7058	.4981	.4015	49.1349	2.0000	99.0000	.0000
<i>Variables</i>						
	<i>Coeff.</i>	<i>SE</i>	<i>t</i>	<i>p</i>	<i>LLCI</i>	<i>ULCI</i>
Constant	.8108	.3377	2.4005	.0182	.1406	1.4809
EWM	.4092	.0948	4.3175	.0000	.2212	.5973
EWA	.4677	.1153	4.0556	.0001	.2389	.6965

Notes: LLCI and ULCI respectively are the lower and upper confidence intervals of the effect coefficients obtained from the sampling distribution generated through bootstrapping.

Table 2 Direct, mediation and total effect of EWA and EWI (continued)

<i>Direct and indirect effects</i>					
<i>Direct effect of EWA on EWI</i>					
<i>Effect</i>	<i>SE</i>	<i>t</i>	<i>p</i>	<i>LLCI</i>	<i>ULCI</i>
.4677	.1153	4.0556	.0001	.2389	.6965
<i>Indirect effect of EWA on EWI</i>					
<i>Effect</i>	<i>Boot SE</i>	<i>BootLLCI</i>		<i>BootULCI</i>	
EWM	.3203	.1053		.1250	
				.5372	

Notes: LLCI and ULCI respectively are the lower and upper confidence intervals of the effect coefficients obtained from the sampling distribution generated through bootstrapping.

Table 2 shows that coefficient of e-waste awareness has a positive relationship with e-waste motivational factor (model 1) and it is significant ($P < 0.05$) thereby supporting Hypothesis 3. R^2 at 0.413 may be called substantial. While we are on the subject, we have also be mentioned here that when we ran the regression, EMI as a function of only awareness of the young users then R^2 comes out to be 0.394. The beta coefficient is 0.787 with standard error 0.095 ($P < 0.05$) lending support to Hypothesis 1. It means that if we acknowledge e-waste motivational factor to the total effect equation, the R^2 of the model goes up significantly. This substantial increase in explained variation lends credence to the hypothesis that e-waste motivation has mediating role in the proposed model. Bhat and Patil (2014) stated that innocence and ignorance of e-product users unintentionally damage the environment, ecology and health of current and future generation. Hence, awareness and motivation towards basic civic sense can help in managing the e-waste among users.

Model 2 shows that EMM and EWI are positively related at 5% level of significance, supporting Hypothesis 2. It implies that motivation towards users' better health, future and sustainable development may induce them to adopt green approach in his/her intended disposal of e-waste. Since the model 2 also shows that there is significant relationship between EWA and EWI, it can be inferred that users' awareness plays a vital role in routing e-waste properly. It may also note that is a modest increase in R^2 of model 2 (0.498) as compared to R^2 of model 1 (0.413). It shows, prima facie, better relevance of model 2 relatively than model1. It implies: one, model 2 has a better model specification compared to model 1 due the modest increase of R^2 . Two, EWA explains greater variation in EWI when EWM also is taken into account. It implies that it not only the awareness which intends to e-waste management, but addition of motivation in it increases the intentions to manage e-waste.

Moreover, there is an indirect effect between the two coefficients namely EWA on EWI through EWM; it is indicated by ab, which was found to be 0.3203 ($P < 0.05$).

It implies that there is a partial mediation effect exists between EWA on EWI through EWM. It may therefore be affirmed that motivation towards better environment and health leads to EWI more in presence of awareness, rather than a mere awareness about the terminology of e-waste. Thus, Hypothesis 4 is supported. Table 3 summarises the support to various hypothesis proposed in this study.

Table 3 Support to different hypotheses

<i>Hypothesis</i>	<i>Proposed effect</i>	<i>Coefficient</i>	<i>Results</i>
H1	+	0.787*	H1 is supported
H2	+	0.409*	H2 is supported
H3	+	0.468*	H3 is supported
H4	+	0.320*	H4 is supported

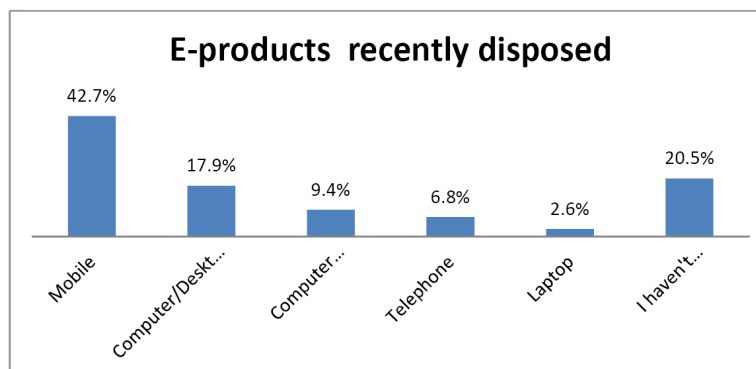
Note: * $P < 0.05$

6 Discussions and conclusions

The present study indicates that young users have awareness about e-waste management. When respondents were asked, whether they are aware about e-waste management, 86% of the respondents stated that they are aware about it, whereas, remaining had very limited and no idea about e-waste management (Figure 3).

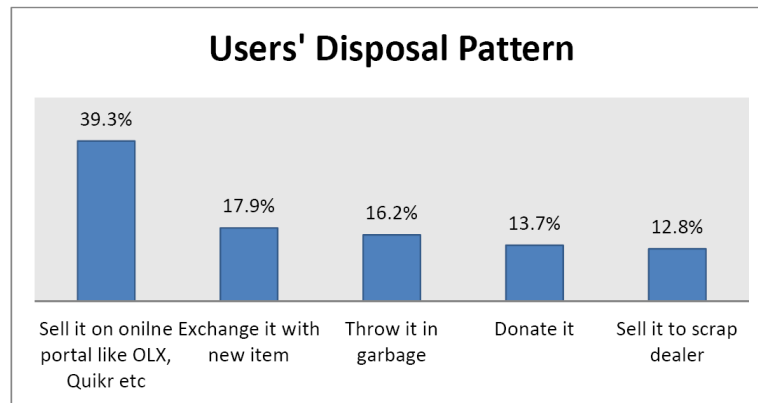
Figure 3 E-waste general awareness (see online version for colours)

When respondents were asked about e-products they have recently disposed, 42.7% disposed their mobile phone followed by computer and desktop. However, there were 20.5% of the respondents who had never disposed their e-products (Figure 4).

Figure 4 Awareness as a function of e-product disposal (see online version for colours)

While probing on how young users dispose their unused e-products, 39.3% of the respondents stated that they dispose their unused e-products on online portal like OLX or Quikr. Moreover, many of the times users exchange the unused products for the new one or donate it. However, 16.2% of the respondents donate their unused e-products. Hence, we can safely assert that young users are initiating towards e-waste management (Figure 5).

Figure 5 E-waste awareness as a function of disposal pattern (see online version for colours)



The survey data and analysis of the present study are conclusive on determination of e-waste management through users' general awareness and motivation. This study aims at providing an approach to comprehend the process of e-waste management intention through general awareness and motivation of the users. The present study empirically explores that general awareness of e-waste influences the intentions to manage the e-waste. In addition, it has been observed that e-waste management intention is partially mediated by the motivation of the users towards better health and environment. All the proposed hypotheses in the present study are supported.

This study illustrates the four major findings. First, the present study proposes an original framework by empirically exploring the relationship among general awareness, EWMs and EWI. Second, young users responded that they are aware about the e-waste management and reported the disposal pattern of unused e-products. Third, there is positive relationship between general e-waste awareness and EWI and this relationship is partially mediated by motivation to route the e-waste for sustainable development. Fourth, this paper extends the research of e-waste management by analysing the impact of awareness and motivation in the field of green IT.

Consequently, environmentally sound e-waste management can help in resource conservation and re-use, material recovery and energy efficiency. Hence, EWA, motivation and intentions have to be articulated in the context of integrated waste management. However, model in the present study needs to be evolved since only the variables present in the current study may not create interest among different stakeholders. Government of India and NGOs need to come with clear and define roles and responsibilities towards e-waste management. Further, integration among all stakeholders is essential to establish workable method for better e-waste management.

7 Limitations and future scope

This study concentrates only on the e-waste management of e-products of young consumers. Future research can focus on the different and other particular age groups to compare the results of this study. Also qualitative research like narrative analysis or interpretative phenomenological approach can be used to capture the experiences of the users to obtain support for positivist approach adopted in this study. Second, this study is based on sample size confined to Delhi and NCR. There is a need to examine the subject over a sample extending to other states of the country. Third, only a few variables of e-waste awareness, motivation and intentions have been examined in the present study. There is a need to investigate other variables also apart from seeking revalidation of the findings. Fourth, factors and dynamics of perceived users towards motivation, awareness and intentions change over period of time. Hence future researchers should consider longitudinal study.

References

- Bandura, A. (1989) 'Regulation of cognitive processes through perceived self-efficacy', *Developmental Psychology*, Vol. 25, No. 5, pp.729–735.
- Barr, S., Gilg, A.W. and Ford, N.J. (2001) 'Differences between household waste reduction, reuse and recycling behaviour: a study of reported behaviours, intentions and explanatory variables', *Environmental and Waste Management*, Vol. 4, No. 2, pp.69–82.
- Bhat, V. and Patil, Y. (2014) 'E-waste consciousness and disposal practices among residents of Pune City', *Procedia – Social and Behavioral Sciences*, Vol. 133, pp.491–498.
- Bui, M.H. (2005) 'Environmental marketing: a model of consumer behavior', *Advances in Marketing*, Vol. 6, No. 2, pp.20–28.
- Goldenhar, L.M. and Connell, C.M. (1993) 'Understanding and predicting recycling behaviour an application of the theory of reasoned action', *Journal of Environmental Systems*, Vol. 22, No. 1, pp.91–103.
- Hicks, C., Dietmar, R. and Eugster, M. (2005) 'The recycling and disposal of electrical and electronic waste in China – legislative and market responses', *Environmental Impact Assessment Review*, Vol. 25, No. 5, pp.459–471.
- Hischier, R., Wäger, P. and Gauglhofer, J. (2005) 'Does WEEE recycling make sense from an environmental perspective?: The environmental impacts of the Swiss take-back and recycling systems for waste electrical and electronic equipment (WEEE)', *Environmental Impact Assessment Review*, Vol. 2, No. 5, pp.525–539.
- Hopper, J.R. and Nielsen, J.M. (1991) 'Recycling as altruistic behavior normative and behavioral strategies to expand participation in a community recycling program', *Environment and Behavior*, Vol. 23, No. 2, pp.195–220.
- Kahhat, R., Kim, J., Xu, M., Allenby, B., Williams, E. and Zhang, P. (2008). 'Exploring e-waste management systems in the United States', *Resources, Conservation and Recycling*, Vol. 52, No. 7, pp.955–964.
- Kang, H.Y. and Schoenung, J.M. (2004). 'Used consumer electronics: a comparative analysis of materials recycling technologies', *Electronics and the Environment, 2004: Conference Record: 2004 IEEE International Symposium*, pp.226–230, IEEE.
- Khetriwal, D.S., Kraeuchi, P. and Widmer, R. (2009) 'Producer responsibility for e-waste management: key issues for consideration-learning from the Swiss experience', *Journal of Environmental Management*, Vol. 90, No. 1, pp.153–165.
- Murugesan, S. (2008) 'Harnessing green IT: principles and practices', *IT Professional*, Vol. 10, No.1, pp.24–33.

- National Youth Policy Report (2014) [online] <http://yas.nic.in/en/yas1/national-youth-policy-2014-document> (accessed 14 July 2015).
- Ni, H.G., Zeng, H., Tao, S. and Zeng, E.Y. (2010) 'Environmental and human exposure to persistent halogenated compounds derived from e-waste in China', *Environmental Toxicology and Chemistry*, Vol. 29, No. 6, pp.1237–1247.
- Nnorom, I.C. and Osibanjo, O. (2008) 'Overview of electronic waste (e-waste) management practices and legislations, and their poor applications in the developing countries', *Resources, Conservation and Recycling*, Vol. 52, No. 6, pp.843–858.
- Preacher, K.J. and Hayes, A.F. (2008) 'Asymptotic and resampling strategies for assessing and comparing indirect effects in multiple mediator models', *Behavior Research Methods*, Vol. 40, No. 3, pp.879–891.
- Preacher, K.J. and Hayes, A.F. (2014) 'Statistical mediation analysis with a multicategorical independent variable', *British Journal of Mathematical and Statistical Psychology*, Vol. 67, No. 3, pp.451–470.
- Rajya Sabha Secretariat, Research Unit (Larrdis), Govt. of India (2011) *E-waste in India* [online] http://rajyasabha.nic.in/rsnew/publication_electronic/E-waste_in_india.pdf (accessed 19 July 2015).
- Robinson, B.H. (2009) 'E-waste: an assessment of global production and environmental impacts', *Science of the Total Environment*, Vol. 408, No. 2, pp.183–191.
- Schultz, P.W., Oskamp, S. and Mainieri, T. (1995) 'Who recycles and when? A review of personal and situational factors', *Journal of Environmental Psychology*, Vol. 15, No. 2, pp.105–121.
- Sharma, N. and Kesharwani, S. (2015) 'Encouraging green purchasing behavior by increasing environmental consciousness', in Das, J.K., Zameer, A., Narula, A. and Tripathi, R. (Eds.): *Reinventing Marketing for Emerging Markets*, pp.288–301, Bloombury Publishing, India.
- Sharma, N. and Sharma, C.S. (2013) 'Encouraging green purchasing behavior through green branding', *Business Analyst*, Vol. 34, No. 2, pp.65–76.
- Simmons, D. and Widmar, R. (1990) 'Motivations and barriers to recycling: toward a strategy for public education', *The Journal of Environmental Education*, Vol. 22, No. 1, pp.13–18.
- Terazono, A., Murakami, S., Abe, N., Inanc, B., Moriguchi, Y., Sakai, S.I. and Williams, E. (2006) 'Current status and research on e-waste issues in Asia', *Journal of Material Cycles and Waste Management*, Vol. 8, No. 1, pp.1–12.
- The Hindu* (2009) 'Rules on e-waste management by March', 20 December.
- Wang, Z., Zhang, B., Yin, J. and Zhang, X. (2011) 'Willingness and behavior towards e-waste recycling for residents in Beijing City, China', *Journal of Cleaner Production*, Vol. 19, No. 9, pp.977–984.

Appendix

EWA	I am aware about the impact of e-waste on human's health and ecology Are you aware about the term e-waste? I am interested in learning about e-waste management I often think about the impact of my actions on natural world?
EWM	I am concerned about e-waste management because of its consequences on my health and welfare I am concerned about e-waste management because of its consequences towards natural environment I am concerned about e-waste management because of the financial benefits out of it
EWI	I will intend to recycle/dispose ICT e-waste I will expect to recycle/dispose ICT e-waste in the future because of its environmental performance and for my own welfare Overall, I will be glad to manage e-waste because it is environmental friendly
